Reg. No. | | | | | | | | | | | [18MTADE203]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

BASICS OF HELICOPTERS

Time: 03 Hours **Maximum Marks:100**

Instructions: Answer five full questions. Question numbers one and two are compulsory. Make suitable assumptions with justifications for any missing data

1. a. With neat sketches, explain the basic principle of
   - i. Helicopter
   - ii. Autogyro and
   - iii. Gyro-dyne **08**

   b. Explain why helicopter is a very complex machine in comparison to aircraft. State and explain the advantages and limitations of helicopter. **08**

   c. Explain co-axial rotor and how anti-torque and directional control is archived **04**

2. a. Briefly explain:
   - i. Induced velocity
   - ii. Disc loading
   - iii. Figure of Merit
   - iv. Autorotation
   - v. Ground Effect. **10**

   b. Describe with a neat sketches, the flow conditions through the main rotor system of a conventional helicopter, under the condition;
   - i. Hover / Vertical Climb
   - ii. Slow descent
   - iii. Zero thrust condition
   - iv. Moderate descent
   - v. Fast descent. State the cases where momentum theory cannot be applied. **10**

3. a. Define solidity ratio, blade loading and average or mean lift coefficient for the helicopter rotor. Show that the average lift coefficient is given by $\overline{C}_L = 6\frac{C_T}{\sigma}$ **10**

   b. A helicopter with 3000 kg hovering at sea level has the following data :
   D= 11 m , c= 0.3 m, tip speed 220 m/s, number of blades = 4, blade section lift curve slope = 5.7 /Rad, C_{D0} of the blade section = 0.01. Calculate;
   - i. C_T, σ, λ
   - ii. Mean lift coefficient $\overline{C}_L$
   - iii. Collective pitch
   - iv. Profile Power
   - v. Inducer Power. **10**

OR

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4. a. Explain geometric twist of the helicopter rotor blade, its requirement and also explain ideal twist. **04**

b. State the assumptions made in momentum theory and which assumptions are overcome in blade element theory. **04**

c. For an ideally twisted blade (θ_t), based on blade element theory, derive the expression for C_T for the rotor in hover $\left[C_T = \frac{a\sigma}{4}(\theta_t - \lambda_i)\right]$ **12**

5. a. Why there is a forward speed limit in a conventional helicopter? Explain with a neat diagram. Can the forward speed be increased by increasing rotor RPM? Explain. **05**

b. With a neat diagram, of the helicopter rotor in forward flight, indicate (i)balance of forces, (ii) Azimuth angle (iii) Advancing side (iv) retreating side (v) Reverse flow region and (vi) High Mach number region. **05**

C. For a helicopter rotor in forward flight, calculate the boundaries of high mach number region (r at $\psi=90^0$, $\psi 1$ and $\psi 2$ at r=R), the radius and position of reverse flow circle for the following data. R=6 m, RPM=330, V= 300KMPH, M_{dd} (Drag divergence Mach number of blade section 0.8 at sea level ISA. Assume speed of sound as 330 m/s **10**

OR

6. a. Show that the radius of the reverse flow region of rotor in forward flight is given by $\frac{\mu R}{2}$. **06**

b. A Helicopter with W=7,400 kg, D=15m, αd=00, U=220 m/s is in level flight at Sea level with forward speed of 100 kmph. Based on momentum theory, calculate the induced velocity (by iteration) for the above condition. **08**

c. With a neat figure of Power required (and power available) Vs Velocity curve for a helicopter explain the salient performance points **06**

7. a. Explain dissymmetry of lift of the rotor in forward flight and its effect. How its effect can be overcome by flapping hinge? Explain. **10**

b. Explain Gyroscopic effect and the rotor response due to gyroscopic effect. **10**

OR

8. a. Show that for the centrally hinged blade in hover, the flapping motion is both statically and dynamically stable. **06**

b. Write the expression in terms of azimuth for pitch and flapping angle of the blade. Explain longitudinal and lateral terms. **08**

c. What are the sources of helicopter vibration? State some of the means of reducing vibration. Explain the principle of passive isolation system. **06**